Modelling the Efficiency of a Sheep Pasture in *Minecraft*

Evan Freeland

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1 Introduction

Minecraft is an open-world sandbox video game[1]. The game takes place in a procedurally generated world made of cubic voxels, or *blocks*. Among other activities, players can engage in agriculture, including the raising of sheep. Sheep can be sheared to gather wool, and must eat grass to regrow their wool after shearing. Grass blocks are converted to dirt after being eaten, and may be converted back to grass by other adjacent grass blocks. If a pasture is packed too densely with sheep, they will deplete the grass, and wool production will be low. However, if the pasture is not densely packed, wool production will still be low because there are not many sheep. Between these two extremes, there is an optimal sheep population that will maximize wool output.

2 Deterministic model

Minecraft operates largely on chance. Both the growth of grass and the appetite of sheep is determined by random number generation. Every time the game's physics engine performs an update, grass has a chance to spread and sheep have a chance to eat. The grass population of the pasture is therefore governed by stochastic processes. Modelling stochastic processes is no simple task, so we will make a first attempt at modelling by assuming the pasture evolves according to deterministic, continuous-time processes.

2.1 Grass mechanics

The *Minecraft* physics engine operates in discrete increments of time, called *ticks*. Every tick, a sheep has a chance to attempt to eat the block it's standing on. If it's standing on grass, it will successfully eat the grass and convert it into dirt. If the sheep and grass in the pasture are uniformly distributed, the expected number of sheep standing on grass is the product of the number of

[1]This investigation was conducted in Minecraft version 1.19.4.

sheep and the fraction of grass blocks in the pasture. The rate at which grass is eaten is therefore given by:

$$\mu = \frac{NRN_g}{A} \tag{1}$$

Where μ is the rate at which grass is eaten, A is the size of the pasture, N_g is the number of grass blocks in the pasture, N is the number of sheep, and R is the frequency with which an individual sheep attempts to eat.

The rate at which grass grows not as simple. If N_g is small, then grass growth will be limited by the lack of places the grass can spread *from*. If N_g is large, grass growth will be limited by a lack of places for the grass to spread *to*. To capture this behavior, we will employ a logistic model, where population follows the differential equation:

$$\dot{y} = ky\left(1 - \frac{y}{p}\right) \tag{2}$$

Where $y(t)$ is the population at time t, k is a parameter representing the reproduction rate of the population and p is the carrying capacity, the maximum population that the local environment can sustain.

The logistic model is one of the oldest models for population growth. It was born from the older exponential (or *Malthusian*) model, but modified to include a maximum population size. It is useful as a tool of instruction, but has since been succeeded by more sophisticated and comprehentive models. However, we are not dealing with real-world ecology, but a video game which has simple population mechanics. Therefore the model is appropriate for our purposes. Let's define λ as the grass growth rate. Our population of interest is the number of grass blocks in the pasture N_g, and the carrying capacity is A:

$$\lambda = kN_g\left(1 - \frac{N_g}{A}\right) \tag{3}$$

Where k is a constant as described above.

The rate of population change is the difference of λ and μ:

$$\begin{aligned}\dot{N}_g = \lambda - \mu &= kN_g\left(1 - \frac{N_g}{A}\right) - \frac{NRN_g}{A} \\ &= \left(k - \frac{NR}{A}\right)N_g\left(1 - \frac{kN_g}{Ak - NR}\right)\end{aligned}$$

Equation (4) defines a logistic model for grass population with the effect of sheep:

$$\dot{N}_g = k^* N_g\left(1 - \frac{N_g}{A^*}\right) \tag{4}$$

Where

$$k^* = k - \frac{NR}{A}$$
$$A^* = A - \frac{NR}{k}$$

N_g will asymptotically approach A^* as time goes on, therefore, we will refer to A^* as the equilibrium grass population.

2.2 Experimental determination of rate constants

Before proceeding further, it's worth determining the actual values of k and R. This will be done via in-game experiments, which will be described briefly here. More detail can be found on the author's github[2].

Ten sheep were locked in cells with a grass block immediately underneath them. The cells included circuitry to count when each sheep ate the grass and automatically replenish it. In one hour, the sheep consumed 320 blocks of grass. This gives an R value of 0.00888 s^{-1}.

A 30x30 patch of dirt was isolated from its surroundings and uniformly interspersed with 100 grass blocks. As the grass grew, the grass population was recorded every 5 seconds. The data was fit to the exact solution of the logistic equation:

$$y = \frac{p}{1 + \left(\frac{p - y_0}{y_0}\right) e^{-kt}} \tag{5}$$

The best fit gave $k = 0.00773$ s^{-1}. The data and fit are compared in Figure 1. The close fit gives us also confidence that the logistic model is appropriate.

2.3 The fast farmer

For simplicity, let's imagine the farmer has superhuman speed, and is able to shear each sheep the instant it grows its wool. In this case, the rate at which wool is gathered is equal to μ. Denoting the rate of wool gathering with F, and substituting $N_g = A^*$ into (1):

$$F_{fast} = \mu = NR - \frac{N^2 R^2}{Ak} \tag{6}$$

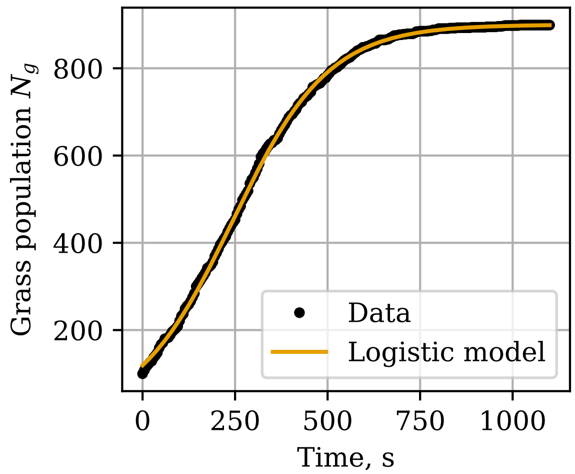
We can find the optimal sheep population by differentiating (6) w.r.t. N, setting the derivative to 0 and solving for N:

$$N = \frac{Ak}{2R} \tag{7}$$

This is the sheep population which will produce the maximum wool output, if the farmer is significantly fast enough. At the optimal value of N, the rate of wool production is:

[2]github.com/ELFREELAND/minecraft_sheep

Figure 1: Logistic model fit to grass growth data.



$$F_{fast} = \frac{Ak}{4}$$

Figure 2 shows F_{fast} and A^* as a function of N, normalized for pasture area. These are simple curves, a parabola and straight line respectively, but there are a few important features to glean. First, A^* and F_{fast} are both zero at $\frac{N}{A} = \frac{k}{R} = 0.869$. Deterministically, the grass in a pasture with $N > 0.869A$ will become extinct.

Second, the maximum F_{fast} coincides with $A^* = 0.5A$. This is a feature of the logistic model for grass growth; population growth is greatest when the population is half of the carrying capacity.

2.4 The slow farmer

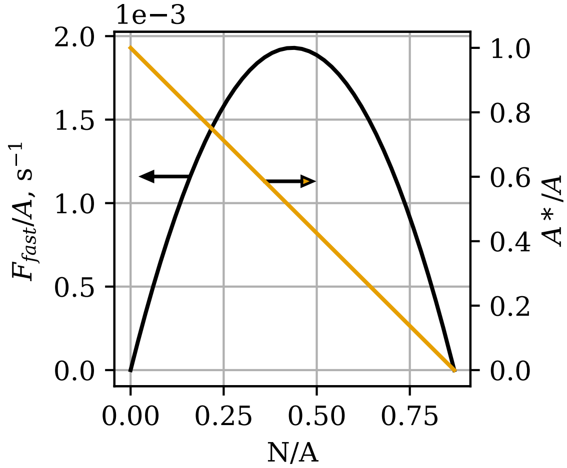
Minecraft players typically don't want to put all of their time into shearing their sheep as quickly as they can. For a more realistic model, we'll assume the farmer shears all sheep at once, intermittently and at regular intervals.

The farmer is off doing other things, and every T seconds, they come back to shear the sheep. Upon shearing the sheep, all the sheep are wool-less. By the time they come back for the next shearing, the sheep have regrown some amount of wool. Let's say the number of sheep that have their wool is N_w. So when the farmer shears the sheep, they get N_w wool. Then the average rate at which the farmer gathers wool is

$$F_{slow} = \frac{N_w}{T}$$

Now we need to consider the value of N_w. A sheep will only regrow wool upon eating if it doesn't already have any, so we should expect that the rate at

Figure 2: Plot of $\frac{F_{fast}}{A}$ and $\frac{A^*}{A}$ as a function of N, normalized to pasture size.



which sheep are regrowing wool is the rate at which they are eating, μ, times the fraction of sheep that are wool-less. We'll assume the pasture has equilibrated and $N_g = A^*$:

$$\dot{N}_w = \frac{N - N_w}{N}\mu = (N - N_w)\frac{RA^*}{A} \tag{8}$$

At time $t = 0$, the farmer shears all the sheep, leaving them with $N_w = 0$, and then at time $t = T$, the farmer comes back and shears them again, gathering N_w blocks of wool. How many sheep grew their wool back in time T? We can find out by solving (8) for N_w. We'll separate it into N_w terms and non N_w terms:

$$\frac{1}{N - N_w}dN_w = \frac{RA^*}{A}dt$$

And integrate from initial state ($t = 0, N_w = 0$) to the final state ($t = T, N_w = N_{w,T}$):

$$\int_0^{N_{w,T}} \frac{1}{N - N_w}dw = \int_0^T \frac{RA^*}{A}dt$$

$$-\ln\left(\frac{N - N_{w,T}}{N}\right) = \frac{TRA^*}{A}$$

And solve for $N_{w,T}$:

$$N_{w,T} = N - Ne^{\frac{TRA^*}{A}}$$

So when the farmer comes back to shear the sheep at $t = T$, they will get $N - N\exp\left(-\frac{TRA^*}{A}\right)$ blocks of wool. If the farmer keeps coming back every T seconds, they will be gathering wool at a rate:

$$F_{slow} = \frac{N - Ne^{-\frac{TRA^*}{A}}}{T}$$

Finally, we can plug in the expression for A^*:

$$F_{slow} = \frac{N - Ne^{-TR+\frac{TR^2N}{Ak}}}{T} \tag{9}$$

This is the expression for the rate of wool gathering by a slow farmer. To find the optimal sheep population, we can take the same approach as we did with the fast farmer case. Differentiating (9) w.r.t. N and setting $F_{slow} = 0$:

$$\frac{dF_{slow}}{dN} = \frac{1 - \left(\frac{TR^2N}{Ak} + 1\right)e^{-TR+\frac{TR^2N}{Ak}}}{T} = 0$$

Solving for N, we obtain the optimal value of N:

$$N = \frac{Ak}{TR^2}(W(e^{TR+1}) - 1) \tag{10}$$

Where W is the Lambert W function, which satisfies $W(xe^x) = x$. Plugging this back into (9), we obtain the value of F_{slow} at optimal N:

$$F_{slow} = \frac{Ak(W(e^{TR+1}) - 1)(1 - e^{-TR+W(e^{TR+1})-1})}{R^2T^2} \tag{11}$$

Figure 3 shows a plot of (9), and figure 4 shows plots of (10) and (11). As T increases, the optimal sheep population also increases, and the optimal rate of wool gathering decreases. Clearly there is a tradeoff between the frequency with which the farmer shears the flock and the rate at which they gather wool.

There are a couple important things to note about F_{slow}. First, compare the plot of F_{fast} in figure 2 with the plot of F_{slow} with $T = 1$ in figure 3 - they are almost identical. We can see this algebraically by employing the following approximation, for $x \ll 1$:

$$e^x \approx x + 1$$

In equation (9), the exponent is small for small values of T, therefore:

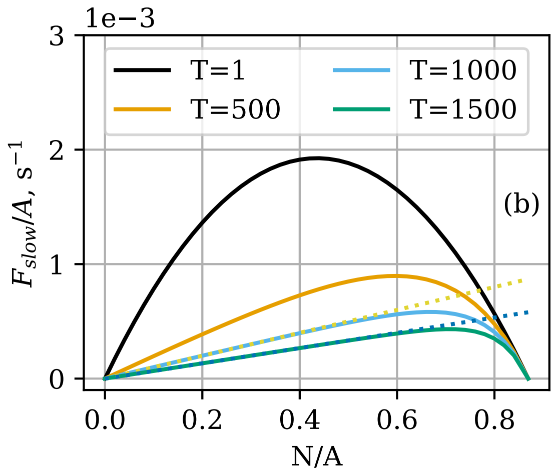
$$F_{slow} \approx \frac{N - N(-TR + \frac{TR^2N}{Ak} + 1)}{T} = NR - \frac{N^2R^2}{Ak} = F_{fast}$$

Second, it appears in figure 3b that for large values of T and modest values of N, F_{slow} is approximately linear w.r.t. N. Under these conditions, the exponent $-TR + \frac{TR^2N}{Ak}$ in (9) is large and negative, and:

$$F_{slow} \approx \frac{N-0}{T} = \frac{N}{T}$$

Physically, this represents conditions for which all of the sheep have regrown their wool by the time the farmer comes back to harvest, so the farmer is simply gathering N blocks of wool with every harvest. Lines of $\frac{N}{T}$ are plotted over top of the curves in figure 3.

Figure 3: Contours of F_{slow} as a function of N at constant T.



3 Stochastic model

The deterministic model gives valuable insight into the expected behavior of the pasture, but a more complete picture can be obtained by developing a stochastic model which captures the randomness inherent in the physics engine of *Minecraft*. We will develop a stochastic model for the grass population in the pasture using Kolmogorov equations. This approach for population modelling is described in more detail in [1], but the key elements will be outlined here.

3.1 Grass Mechanics

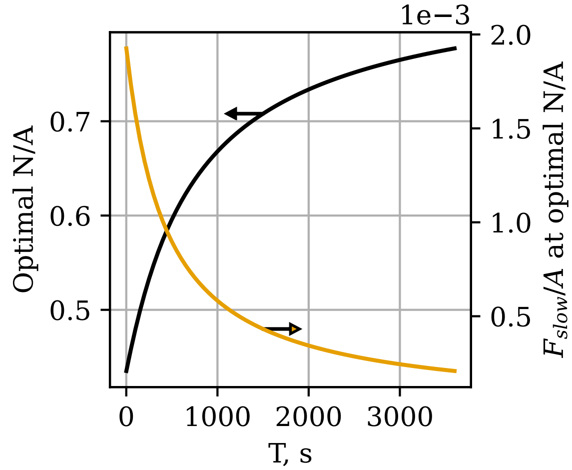
In the deterministic model, we considered the grass population as a single number, N_g, which evolved with time. In the stochastic approach, we will consider the evolution of the *probability distribution* of N_g with time.

We'll introduce the following notation:

$$p_x(t) = \Pr(N_g(t) = x)$$

In words, $p_x(t)$ is the probability that the pasture has $N_g = x$ at time t.

Figure 4: Optimal sheep population and maximum wool production as a function of T, normalized to pasture size.



We'll also need to slightly reconsider the meanings of μ and λ. In the deterministic model, these are the rate of grass growth or consumption. For a small time interval Δt, the number of grass blocks that are eaten is $\mu\Delta t$, and likewise for λ. If Δt is sufficiently short so that the probability of two independent changes in grass population is negligible, $\mu\Delta t$ can be reinterpreted as the probability of one grass block being consumed in time Δt, and likewise for λ. μ and λ therefore represent the probability of the pasture transitioning to a state with one more or one less grass block than before.

Using these new interpretations of μ and λ, we can write a system of differential equations describing the evolution of the probability distribution of N_g with time. Consider a pasture at time $t = t_0$, with $N_g(t_0) = x$. A short time ago, at time $t = t_0 - \Delta t$, the pasture had $N_g(t_0 - \Delta t)$. The pasture may have arrived at $N_g(t_0) = x$ in one of four ways:

1. $N_g(t_0 - \Delta t) = x$, and neither eating nor growth take place.
2. $N_g(t_0 - \Delta t) = x - 1$, and one grass block regrows.
3. $N_g(t_0 - \Delta t) = x + 1$, and one grass block is eaten.
4. Two or more growth and eating events occur.

The probability of each pathway is the probability that the pasture was in the required state $N_g(t_0 - \Delta t)$ times the probability of the event (eating or growth) required to bring the pasture to $N_g(t_0) = x$. The probability of one grass growth in time Δt is $\lambda\Delta t$, and the probability of one grass block being eaten is $\mu\Delta t$. This gives the probability of nothing happening as $(1 - \lambda - \mu)\Delta t$. Finally, the probability of multiple events is some product of $\lambda\Delta t$ and $\mu\Delta t$, and is therefore dependent on Δt with order 2 or higher. The probability of each path is therefore:

$$\begin{aligned}
\Pr(1) =& p_x(t_0 - \Delta t)(1 - \lambda - \mu)\Delta t \\
\Pr(2) =& p_{x-1}(t_0 - \Delta t)\lambda\Delta t \\
\Pr(3) =& p_{x+1}(t_0 - \Delta t)\mu\Delta t \\
\Pr(4) \approx& O(\Delta t^2)
\end{aligned}$$

The probability of the pasture arriving at $N_g(t_0) = x$ is the sum of the probability of each pathway:

$$\begin{aligned}
p_x(t_0) = & \; p_x(t_0 - \Delta t)(1 - \lambda - \mu)\Delta t \\
& + p_{x-1}(t_0 - \Delta t)\lambda\Delta t \\
& + p_{x+1}(t_0 - \Delta t)\mu\Delta t \\
& + O(\Delta t^2)
\end{aligned}$$

subtracting $p_x(t_0 - \Delta t)$ and dividing by Δt:

$$\begin{aligned}
\frac{p_x(t_0) - p_x(t_0 - \Delta t)}{\Delta t} = & -p_x(t_0 - \Delta t)(\lambda + \mu) \\
& + p_{x-1}(t_0 - \Delta t)\lambda \\
& + p_{x+1}(t_0 - \Delta t)\mu \\
& + O(\Delta t^1)
\end{aligned}$$

and taking the limit as $\Delta t \to 0$:

$$\dot{p}_x(t) = -p_x(t)(\lambda(x) + \mu(x)) + p_{x-1}(t)\lambda(x-1) + p_{x+1}(t)\mu(x+1) \tag{12}$$

Note that $\mu(x)$ simply represnts the value of μ with $N_g = x$, and likewise for λ. Note also that the $O(\Delta t^2)$ term has vanished upon taking the limit $\Delta t \to 0$.

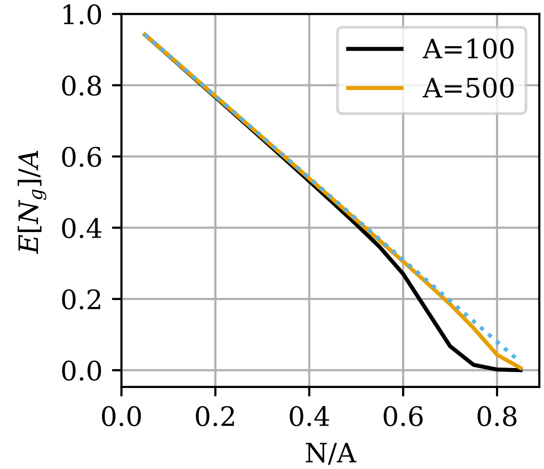
Systems of equations such as these, which describe the evolution of a probability distribution with time, are known as *Kolmogorov equations*. We will have a system of $A + 1$ equations, one for each possible value of N_g.

We'll wrap up the system into vector form for convenient expression. Let $\mathbf{p}$ be a vector of p_x, and $\dot{\mathbf{p}}$ its derivative:

$$\mathbf{p}(t) = [p_0(t), p_1(t), p_2(t) \ldots p_A(t)]$$

$$\dot{\mathbf{p}}(t) = \frac{d\mathbf{p}}{dt} = [\dot{p}_0(t), \dot{p}_1(t), \dot{p}_2(t) \ldots \dot{p}_A(t)]$$

Figure 5: Expected value of N_g at $t = 10000$, for two values of A. A^* from the deterministic model is shown as a dotted line.



Let $\mathbf{R}$ be a matrix of coefficients relating $\mathbf{p}$ and $\dot{\mathbf{p}}$:

$$\mathbf{R} = \begin{bmatrix} r_{0,0} & r_{0,1} & r_{0,2} & \cdots & r_{0,A} \\ r_{1,0} & r_{1,1} & r_{1,2} & \cdots & r_{1,A} \\ r_{2,0} & r_{2,1} & r_{2,2} & \cdots & r_{2,A} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{A,0} & r_{A,1} & r_{A,2} & \cdots & r_{A,A} \end{bmatrix}$$

The coefficient $r_{i,j}$ relates p_i with $\dot{p}_j$:

$$\dot{p}_j = \sum_i r_{ij} p_i$$

$$r_{i,j} = \begin{cases} r_{j-1,j} = \lambda(j-1) \\ r_{j+1,j} = \mu(j) \\ r_{i,i} = -(\lambda(i) + \mu(i)) \\ 0 \text{ otherwise} \end{cases}$$

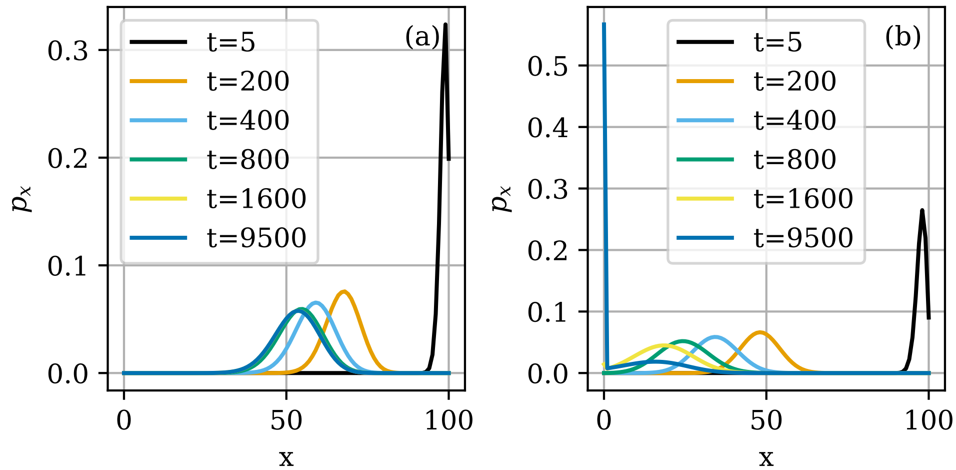
Then:

$$\dot{\mathbf{p}}(t) = \mathbf{p}(t)\mathbf{R}$$

All that's left to do now is to pick values for N and A and an initial value for N_g and solve. This is most easily done numerically, as has been done here using the Runge-Kutta method.

We can already see the stochastic model deviating from the deterministic model. At values of N close to $\frac{Ak}{R}$, the expected grass population density is lower than predicted by the deterministic model.

Figure 6: Evolution of N_g distribution for a pasture of size $A = 100$, for (a) $N = 40$ and (b) $N = 70$.



Let's look in a little more detail at the behavior of the pasture. Figure 6 shows the evolution of the distribution of N_g for $N = 40$ and $N = 70$ in a 100-block pasture. Both pastures begin with $N_g = A$, i.e. $\mathbf{p} = [0, 0, 0, \ldots 0, 1]$. The pasture with $N = 40$ approaches a stable state, evidenced by the indistinguishable distributions at $t = 1600$ s and $t = 9500$ s. In contrast, the pasture with $N = 70$ continues evolving toward smaller and smaller grass populations, with p_0 increasing dramatically as time proceeds.

The reason for this isn't hard to see by considering the equation for $\dot{p}_0$. Beginning with (12) and considering that $p_{-1} = 0$ and $\lambda(0) = \mu(0) = 0$:

$$\dot{p}_0(t) = p_1(t)\mu(1)$$

Since p_1 and $\mu(1)$ are always positive, p_0 is monotonically increasing, bounded above by $p_0 = 1$. Therefore the true equilibrium distribution is $\mathbf{p} = [1, 0, 0, 0, \ldots]$ i.e. the grass has become exinct. However, even if this is the equilibrium state, the pasture may remain at a different *metastable* state if the probability of extinction remains negligible across reasonable time scales. This is what is seen in figure 6a. The distribution for $N = 40$ "stabilizes" with a value of p_1 so low that $\dot{p}_0$ remains negligible. The metastable distribution can be also approximated analyically [3] (see also [1] section 6.8.2).

Rapid grass extinction at high sheep densities poses a problem, since the slow farmer deterministic model predicts that the optimal N increases as T increases (see figure 4). Fortunately, an exact solution can be obtained for the expected time to extinction ETE. Let $\mathbf{M}$ be a matrix of mean residence times, such that the element m_{ij} of $\mathbf{M}$ is the expected amount of time the pasture spends with $N_g = j$ before reaching extinction, given that it began with a population i. It can be shown [2] that $\mathbf{M}$ is obtained by removing the first row and column from

$\mathbf{R}$ and taking the negative of the inverse of this modified $\mathbf{R}$ matrix. Denoting the modified $\mathbf{R}$ matrix $\mathbf{R}^{\dagger}$:

$$\mathbf{M} = -(\mathbf{R}^{\dagger})^{-1}$$

The sum of the ith row of $\mathbf{M}$ gives the total time spent with nonzero population before becoming extinct, given the pasture started with $N_g = j$. The typical *Minecraft* pasture begins fully populated with grass, i.e. $N_g(0) = A$. Therefore the value we are interested in is:

$$\text{ETE} = \sum_j m_{Aj}$$

We can apply this solution to obtain the ETE for any value of N and A, as demonstrated in figure 7. For populations close to the deterministic sheep population limit $\frac{N}{A} = \frac{k}{R} = 0.869$, the ETE is on the order of a few hours for all three pasture sizes. As the sheep population density decreases, the ETE increases dramatically - For a 100 block pasture with a sheep density of 0.5, the ETE is on the order of several years. ETE also increases dramatically with larger pastures - A 100-block pasture with sheep density 0.7 has ETE on the order of a few hours, while a 500-block pasture with the same density has ETE on the order of months.

Let's now turn our attention to the metastable, or quasi-equilibrium, state of the pasture. The pasture will cease evolving when all $\dot{p}_x = 0$. This implies that the likelihood of the pasture having $N_g = x$ and losing one grass block equals the probability of $N_g = x - 1$ and growing one grass block [3]. In symbols:

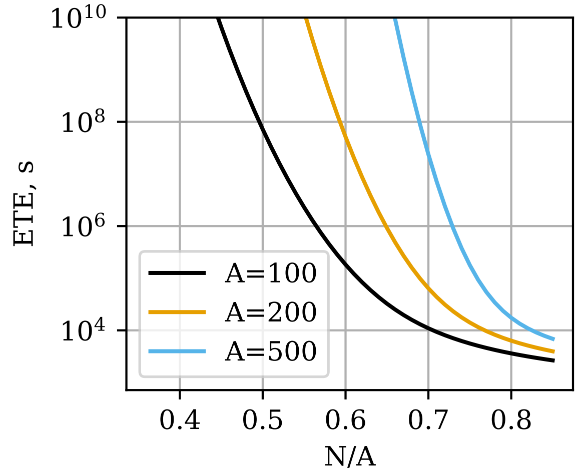
$$p_x \mu(x) = p_{x-1} \lambda(x-1) \tag{13}$$

We will assume $N_g = 0$ is an unreachable state and exclude it from consideration. Writing (13) for $2 \leq x \leq A$ and combining with the normalization condition, we obtain a system of A equations in A variables:

$$\begin{cases} p_2 \mu(2) - p_1 \lambda(1) = 0 \\ p_3 \mu(3) - p_2 \lambda(2) = 0 \\ \dots \\ p_A \mu(A) = p_{A-1} \lambda(A-1) = 0 \\ \sum_{x=1}^{A} p_x = 1 \end{cases} \tag{14}$$

We can solve (14) to obtain the metastable N_g distribution. Figure **??** shows the metastable distribution for a few select conditions. There are a few important features to note. First, figure **??** shows both the metastable distribution predicted by (14) and numerical solutions of the Kolmogorov equations for $t = 10000$. The results match in each case except for the case $A = 100$, $N = 65$. In that case, the numerical solution has already begin to approach extinction. This is the only case in which the metastable p_1 is significant, and it is also the only case in which p_0 is significant in the numerical solution. Intuitively,

Figure 7: Expected time to extinction for pastures of $A = 100$, 200, and 500 as a function of sheep population density.



one might gather that for conditions where the metastable p_1 is significant, the pasture will not remain in the metastable state for long as p_0 will rapidly increase.

3.2 The fast farmer

3.3 The slow farmer

It seems like a natural next step to pursue an analysis of the slow farmer situation with Kolmogorov equations. However, this analysis is not so simple. We now have two variables evolving at the same time, N_g and N_w. Therefore, we will need to consider a bivariate probability distribution. As before, we will denote this as $p_{x,y}(t)$:

$$p_{x,y}(t) = \Pr(N_g(t) = x, N_w(t) = y)$$

Consider a pasture at time t_0, with $N_g(t_0) = x$ and $N_w(t_0) = y$. A short time ago, at time $t = t_0 - \Delta t$, the pasture had $N_g(t_0 - \Delta t)$ and $N_w(t_0 - \Delta t)$. The pasture may have arrived at the state at t_0 in one of several ways:

- $N_g(t_0 - \Delta t) = N_g(t_0)$ and $N_w(t_0 - \Delta t) = N_w(t_0)$, and neither eating nor growth take place.
- $N_g(t_0 - \Delta t) = N_g(t_0) - 1$ and $N_w(t_0 - \Delta t) = N_w(t_0)$, and one grass block regrows.
- $N_g(t_0 - \Delta t) = N_g(t_0) + 1$ and one grass block is eaten:

- $N_w(t_0 - \Delta t) = N_w(t_0)$, and the sheep that ate the grass already had its wool.
- $N_w(t_0 - \Delta t) = N_w(t_0) - 1$, and the sheep that ate the grass did not previously have wool.

- Two or more growth and eating events occur.

As with the fast-farmer model, the probability of each path is the probability of the pasture existing at the required initial state, times the probability of the required change occuring. For the two cases involving one block of grass being eaten, the probability of the event will include a term pertaining to the number of sheep who have their wool. The probability of the pasture arriving at $N_g = x$ and $N_w = y$ is the sum of the probabilities of each path:

$$\begin{aligned} p_{x,y}(t) = {} & p_{x,y}(t - \Delta t)(1 - \mu(x) - \lambda(x))\Delta t \\ & + p_{x-1,y}(t - \Delta t)\lambda(x-1)\Delta t \\ & + p_{x+1,y}(t - \Delta t)\mu(x+1)\frac{y}{N}\Delta t \\ & + p_{x+1,y-1}(t - \Delta t)\mu(x+1)\frac{N-(y-1)}{N}\Delta t \\ & + O(\Delta t^2) \end{aligned}$$

Subtracting $p_{x,y}(t - \Delta t)$, dividing by Δt, and taking the limit $\Delta t \to 0$:

$$\begin{aligned} \dot{p}_{x,y}(t) = {} & p_{x,y}(t)(-\mu(x) - \lambda(x)) \\ & + p_{x-1,y}(t)\lambda(x-1) \\ & + p_{x+1,y}(t)\mu(x+1)\frac{y}{N} \\ & + p_{x+1,y-1}(t)\mu(x+1)\frac{N-(y-1)}{N} \end{aligned} \tag{15}$$

We will have one equation for each combination of N_g and N_w, i.e. $(N+1)(A+1)$ equations. It's tempting to proceed by coming up with a matrix of $p_{x,y}$ and some tensor of coefficients relating them, but let's step back and remember that we are going to be solving this system numerically. A pasture of $A = 200$ and $N = 100$ will require a system of 20301 equations. In principle, a system of simultaneous differential equations of any size can be solved numerically with enough processor time, but in practice, a system of this size will take inconveniently long to solve on an average personal computer.

Recall that when we were analyzing the deterministic slow farmer model, we assumed that the pasture had already reached equilibrium. We saw in the previous section that pastures with long ETE will assume a metastable distribution. Therefore we will try to simplify this system of equations by assuming the pasture has reached a metastable distribution.

$p_{x,y}$ is the probability of the joint event of $N_g = x$ and $N_w = y$. Assuning the events are independent, we can rewrite $p_{x,y}$ as:

$$p_{x,y} = p_x q_y$$

Where $p_x = \Pr(N_g = x)$ and $q_y = \Pr(N_w = y)$.

We can rewrite (15) with this in mind:

$$\begin{aligned}
\dot{p}_{x,y}(t) = p_x \dot{q}_y(t) = & \, p_x q_y(t)(-\mu(x) - \lambda(x)) \\
& + p_{x-1} q_y(t) \lambda(x-1) \\
& + p_{x+1} q_y(t) \mu(x+1) \frac{y}{N} \\
& + p_{x+1} q_{y-1}(t) \mu(x+1) \frac{N-(y-1)}{N}
\end{aligned}$$

We'll assume p_x is time-independent, so $\frac{d(p_x q_y)}{dt} = p_x \frac{dq_y}{dt}$.

Summing $p_{x,y}$ over all x gives q_y, therefore, summing $\dot{p}_{x,y}$ over all x gives $\dot{q}_y$:

$$\begin{aligned}
\dot{q}_y(t) = \sum_{x=0}^{A} \dot{p}_{x,y} = \sum_{x=0}^{A} [& p_x q_y(t)(-\mu(x) - \lambda(x)) \\
& + p_{x-1} q_y(t) \lambda(x-1) \\
& + p_{x+1} q_y(t) \mu(x+1) \frac{y}{N} \\
& + p_{x+1} q_{y-1}(t) \mu(x+1) \frac{N-(y-1)}{N}]
\end{aligned} \tag{16}$$

By summing across all x, we have reduced our system of equations to size $N+1$. We can express our system in vector form, as before:

$$\mathbf{q}(t) = [q_0(t), q_1(t), q_2(t), \ldots q_N(t)]$$

$$\dot{\mathbf{q}}(t) = \mathbf{q}(t)\mathbf{R}$$

The coefficient $r_{i,j}$ relates q_i with $\dot{q}_j$:

$$r_{i,j} = \begin{cases} r_{j,j} = \sum_{x=0}^{A} [p_x(-\mu(x) - \lambda(x)) + p_{x-1}\lambda(x-1) + p_{x+1}\mu(x+1)\frac{j}{N}] \\ r_{j-1,j} = \sum_{x=0}^{A} p_{x+1}\mu(x+1)\frac{N-(j-1)}{N} \end{cases}$$

$$r_{i,j} = \begin{cases} r_{j,j} = \sum_{x=0}^{A} [-p_x\mu(x) - p_x\lambda(x) + p_{x-1}\lambda(x-1) + p_{x+1}\mu(x+1)\frac{j}{N}] \\ r_{j-1,j} = \sum_{x=0}^{A} p_{x+1}\mu(x+1)\frac{N-(j-1)}{N} \end{cases}$$

$$r_{i,j} = \begin{cases} r_{j,j} = \sum_{x=0}^{A} [-p_x\lambda(x) + p_{x+1}\mu(x+1)\frac{j}{N}] \\ r_{j-1,j} = \sum_{x=0}^{A} p_{x+1}\mu(x+1)\frac{N-(j-1)}{N} \end{cases}$$

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